

**COURSE: CHE 312: Transport Phenomena II**

**Fall Semester 2025**

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(65)

**Quiz #4**

**November 18, 2025**

***Open Notes and Book***

**Duration: 40 minutes**

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Consider the irreversible conversion of species A into B and F through the reactions  $A \rightarrow B$  and  $A \rightarrow F$  in a stationary liquid film of thickness  $L$ . Determine  $C_A$  throughout the film, if the surface at  $y = L$  is assumed to be impermeable and unreactive, and the concentration of A at  $y = 0$  is specified as  $C_0$ . The homogeneous reactions follow zeroth and first order kinetics with rates of reaction  $R_{AB} = -k_B$  and  $R_{AF} = -k_F \cdot C_A$ , respectively. The diffusivity of the species A is  $D_A$ .

Find the overall rate of reactant consumption.

$$K = -K_B - K_F \cdot C_A$$

Balance Equation:

$$\frac{d}{dy} \left( D_A \frac{dC_A}{dy} \right) = K$$

Boundary Conditions:

$$\text{B.C. 1: } C_A(y=0) = C_0$$

$$\text{B.C. 2: } \left. \frac{dC_A}{dy} \right|_{y=L} = 0$$

$$N_A(y=0) = -D_A \frac{dC_A(y=0)}{dy}$$

$$D_A \frac{d^2 C_A}{dy^2} = -K_F \left( C_A - \frac{K_B}{K_F} \right)$$

$$N_A(y=0) = \int_0^L (K_B + K_F C_A(y)) dy$$

top boundary from B.C.2

for  $C_A(y)$  use  $\cosh(\lambda)$  where  $\lambda = \sqrt{\frac{K_F}{D_A}}$

B.C.1:

$$C_A(y) = \left( C_0 + \frac{K_B}{K_F} \right) \cosh \sqrt{\frac{K_F}{D_A}} y - \frac{K_B}{K_F} \quad ??$$

$$D_A \frac{d^2 C_A}{dy^2} = K$$

$$N_A(y=0) = -\frac{e^0}{K}$$

$$\frac{D_A}{K} \frac{d^2 C_A}{dy^2} = 1$$

$$D_A \ln K \frac{dC_A}{dy} = y + A$$

$$D_A \frac{dC_A}{dy} = \frac{e^{y+A}}{K}$$

$$\text{B.C. 2: } 0 = \frac{e^{L+A}}{K} \rightarrow A=0$$